Reinforcement Learning–Based Traffic Signal Control with Network-Aware Edge Sensing

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***Abstract*—Urban traffic congestion at signalized intersections has become a critical concern due to rapid urbanization and increasing vehicular density. Conventional traffic signal control strategies, including fixed-time and rule-based approaches, lack adaptability to dynamic traffic patterns and often lead to excessive waiting times and inefficient signal utilization. Recent advances in reinforcement learning (RL), particularly Deep Q-Networks (DQN), multi-agent reinforcement learning, and hierarchical learning frameworks, have demonstrated promising results in adaptive traffic signal optimization. However, most existing RL-based systems assume ideal sensing conditions and delay-free communication between traffic monitoring units and control systems, which is unrealistic in practical intelligent transportation deployments.**

This paper proposes a network-aware reinforcement learning framework for traffic signal control that integrates edge-simulated vehicle detection and communication delay modeling within the learning process. Vehicle density is extracted from recorded traffic videos using the YOLOv8 object detection algorithm and transmitted through a simulated network layer introducing controllable latency and packet loss. A DQN-based controller, implemented in the SUMO simulation environment via the TraCI interface, learns optimal signal phase decisions under varying traffic and network conditions. Experimental evaluation using average waiting time, queue length, and throughput demonstrates improved efficiency and robustness compared to traditional controllers under realistic communication constraints.

***Index Terms*—Reinforcement Learning, Deep Q-Network, Traffic Signal Control, YOLOv8, SUMO, Communication Delay, Intelligent Transportation Systems, Edge Computing**

I. Introduction

Urban traffic congestion has become a significant challenge in modern transportation systems due to rapid urbanization, increasing vehicle ownership, and limited infrastructure expansion. Signalized intersections represent critical control points within urban road networks, where inefficient signal timing strategies directly contribute to long vehicle queues, increased travel delay, fuel consumption, and environmental pollution. Traditional traffic signal control systems, including fixed-time scheduling and rule-based approaches, operate using predefined timing plans and heuristic rules that lack adaptability to dynamic and stochastic traffic conditions. As highlighted in recent comprehensive surveys on reinforcement learning (RL)-based traffic signal control, conventional methods fail to efficiently utilize real-time traffic information and often result in suboptimal intersection performance [1], [2].

In recent years, reinforcement learning and deep reinforcement learning (DRL) have emerged as promising solutions for adaptive traffic signal optimization. Value-based approaches such as Deep Q-Networks (DQN) and policy-based methods have demonstrated significant improvements in reducing average waiting time, queue length, and overall congestion in simulation environments [9], [11]. Scalable DRL frameworks have been proposed to manage large-scale urban intersections, showing superior adaptability compared to fixed-time controllers [4]. Furthermore, multi-agent reinforcement learning (MARL) strategies enable coordination among neighboring intersections, enhancing global traffic efficiency in complex urban networks [5], [6], [15]. Hierarchical reinforcement learning architectures have also been introduced to decompose traffic control tasks into multiple levels, improving learning stability and scalability [3].

Despite these advancements, most existing RL-based traffic signal control systems assume ideal sensing environments and instantaneous communication between traffic monitoring units and centralized controllers. In practical intelligent transportation systems, traffic data is typically collected using distributed edge devices such as cameras and IoT sensors, which process visual information locally before transmitting summarized traffic states over communication networks. Communication latency, bandwidth limitations, and packet loss can significantly affect the timeliness and reliability of control decisions. Robustness studies have indicated that RL policies trained under idealized conditions may degrade when

exposed to real-world uncertainties and disruptions [8]. Recent research also emphasizes the integration of edge computing and communication-aware optimization in intelligent transportation frameworks [12], [13], yet limited work explicitly incorporates network delay modeling into the reinforcement learning training process.

Additionally, emerging paradigms such as federated deep reinforcement learning aim to enable distributed learning across intersections while preserving privacy and scalability [7]. While these approaches improve coordination and deployment feasibility, explicit modeling of communication constraints during policy learning remains underexplored. Consequently, there exists a research gap in designing RL-based traffic signal control systems that simultaneously consider adaptive optimization, edge-based perception, and realistic communication limitations.

To address this gap, this paper proposes a network-aware reinforcement learning framework for adaptive traffic signal control that integrates edge-simulated vehicle detection with communication delay modeling. Vehicle density information is extracted using a YOLOv8-based perception module and transmitted through a simulated communication layer introducing controllable latency and packet loss. A DQN-based controller implemented in the SUMO traffic simulation environment learns optimal signal phase policies under varying traffic and network conditions. By embedding communication constraints within the learning process, the proposed approach aims to enhance robustness and provide a more realistic foundation for deployable intelligent traffic management systems.

II. Literature Survey

Reinforcement learning (RL) has emerged as a powerful approach for adaptive traffic signal control due to its capability to learn optimal policies directly from interaction with traffic environments. Recent surveys have extensively analyzed the evolution of RL-based traffic control systems, categorizing them into value-based, policy-based, and hybrid deep reinforcement learning (DRL) frameworks. Michailidis et al. [1] provide a comprehensive review of RL applications in traffic signal control, highlighting improvements in delay reduction, queue minimization, and adaptive phase timing. Similarly, Saadi et al. [2] survey deep reinforcement learning approaches for intelligent traffic systems and emphasize the importance of state representation, reward design, and scalability in practical deployments.

Among value-based methods, Deep Q-Networks (DQN) have been widely adopted for discrete signal phase control problems. Bouktif et al. [9] demonstrated that careful design of state and reward functions significantly enhances learning stability and performance in DQN-based traffic signal control systems. Zhao and Zhang [11] implemented a DQN-based adaptive traffic signal controller within the SUMO simulation environment and reported substantial reductions in average waiting time compared to traditional fixed-time controllers. These studies confirm the effectiveness of DQN in handling intersection-level optimization tasks.

To address scalability issues in urban traffic networks, multi-agent reinforcement learning (MARL) has gained attention. Shi [5] presented a survey on MARL-based traffic signal control, discussing coordination strategies, communication mechanisms, and reward-sharing approaches for improving network-level performance. Olusanya et al. [6] proposed a cooperative multi-agent framework that enables intersections to share local traffic information, resulting in improved global traffic flow and congestion reduction. Wang and Xu [15] further demonstrated that coordinated multi-agent deep reinforcement learning enhances performance in large-scale traffic networks by mitigating congestion propagation between intersections.

Hierarchical reinforcement learning has also been explored to decompose complex traffic control problems into manageable sub-tasks. Shen et al. [3] proposed a hierarchical RL-based traffic signal control strategy that separates high-level decision-making from low-level signal timing control. Their results indicate improved learning efficiency and adaptability under varying traffic patterns. Similarly, Bao et al. [4] introduced a scalable deep reinforcement learning framework capable of optimizing signal timing across multiple intersections, demonstrating significant performance gains in simulated urban scenarios.

Recent research has also investigated decentralized and distributed learning approaches to improve scalability and privacy. Li et al. [7] proposed a federated deep reinforcement learning framework for urban traffic signal optimization, enabling multiple intersections to collaboratively train models without sharing raw traffic data. This approach addresses data privacy concerns while maintaining adaptive learning capabilities.

While significant progress has been achieved in adaptive optimization and coordination strategies, robustness under uncertain and realistic conditions remains a critical concern. Nguyen et al. [8] examined the robustness of reinforcement learning-based traffic signal control under incident conditions and unexpected disruptions. Their findings reveal that RL policies trained under ideal simulation environments may experience performance degradation when exposed to environmental variations. This observation underscores the importance of incorporating uncertainty modeling and robustness evaluation into traffic signal learning frameworks.

Furthermore, recent advancements in intelligent transportation systems emphasize the integration of edge computing and communication-aware control mechanisms. Although several studies explore edge-based sensing and distributed optimization [7], explicit modeling of communication latency and packet loss within reinforcement learning traffic control remains limited. Most RL-based traffic control studies assume instantaneous and accurate state information from sensors, which may not reflect real-world deployments where network constraints influence system performance.

In summary, the literature demonstrates that reinforcement learning, particularly DQN, hierarchical RL, and multi-agent frameworks, significantly improves traffic signal control performance compared to traditional methods [1]– [4], [9].

However, despite advancements in scalability, coordination, and distributed learning [7], [15], limited attention has been given to incorporating realistic communication constraints and robustness considerations during training [8]. This gap motivates the development of a network-aware reinforcement learning framework that integrates edge-based perception and communication delay modeling within the adaptive control loop, thereby enhancing realism and deployment feasibility in intelligent transportation systems.

III. Methodology and Architecture

The proposed research presents a network-aware reinforcement learning framework for adaptive traffic signal control that integrates edge-based perception and communication delay modeling within the decision-making process. While previous reinforcement learning approaches have demonstrated significant improvements in traffic signal optimization [1], [2], most studies assume idealized simulation conditions where traffic state information is perfectly and instantaneously available to the learning agent. However, practical intelligent transportation systems rely on distributed sensing infrastructure and networked communication channels, where latency and transmission uncertainties may affect control stability. To address this limitation, the proposed methodology explicitly incorporates edge-simulated vehicle detection and communication delay modeling into the reinforcement learning training and evaluation pipeline.

The overall system architecture consists of three interconnected layers: an edge perception layer, a communication layer, and a centralized reinforcement learning control layer interacting with the SUMO traffic simulation environment. The perception layer emulates real-world traffic monitoring by processing recorded traffic videos using the YOLOv8 object detection algorithm integrated with OpenCV. Similar vision-based traffic state extraction mechanisms have been explored in recent intelligent transportation research [3]. The perception module detects vehicles and computes lane-wise vehicle counts at predefined time intervals, generating compact traffic state vectors. This approach ensures modularity and reflects practical deployments where edge devices preprocess visual data before transmitting summarized traffic information.

The processed traffic state is transmitted to a centralized reinforcement learning controller through a simulated communication layer. Unlike traditional RL traffic control systems that assume instantaneous state updates, the proposed framework introduces artificial communication latency and packet loss to emulate realistic network conditions. The communication module is implemented using lightweight protocols such as MQTT or REST-based messaging. Different latency scenarios are considered, including ideal (no delay), moderate delay, and high-delay conditions. Robustness concerns highlighted in recent studies [8] motivate the need to evaluate control policies under such uncertainties. By exposing the reinforcement learning agent to delayed state information during training, the proposed approach enhances policy resilience against communication disruptions.

Traffic signal control is modeled as a sequential decision-making problem where the agent interacts with the environment to optimize long-term traffic efficiency. At each decision step, the agent observes a state representation composed of lane-wise vehicle counts, current signal phase, and optional indicators of information staleness due to communication delay. The action space consists of discrete signal phase transitions, including maintaining the current phase or switching to a permissible next phase. This formulation aligns with value-based traffic signal control frameworks discussed in prior work [9], [11].

A Deep Q-Network (DQN) algorithm is employed as the learning mechanism due to its suitability for discrete action optimization and demonstrated effectiveness in traffic signal control applications [4], [9]. The DQN approximates the action-value function through a neural network trained using experience replay and target network stabilization techniques. Exploration during training is achieved using an ε-greedy strategy to balance exploration and exploitation. The reinforcement learning agent interacts with the SUMO simulation environment through the TraCI interface, allowing real-time retrieval of traffic metrics and application of signal control decisions. SUMO has been widely adopted in recent reinforcement learning traffic research due to its flexibility and microscopic traffic modeling capabilities [4], [11].

The reward function is designed to minimize congestion by penalizing average vehicle waiting time and total queue length while discouraging excessive signal switching. Similar reward structures have been shown to improve convergence and traffic efficiency in scalable RL-based signal optimization frameworks [1], [4]. The integration of network delay modeling into the reward evaluation further enables the agent to learn policies that maintain stable performance under imperfect information conditions.

Compared to hierarchical reinforcement learning strategies [3], multi-agent coordination frameworks [6], [15], and federated learning approaches [7], the proposed architecture specifically focuses on embedding communication awareness within the adaptive control loop. While existing research improves scalability and coordination, explicit modeling of network-induced delays during training remains limited. By combining edge-based perception with delay-aware reinforcement learning, the proposed methodology advances toward realistic and deployable intelligent traffic signal control systems.

Overall, the architecture ensures modularity, scalability, and robustness. The perception, communication, and control layers operate cohesively to emulate real-world intelligent transportation infrastructure while maintaining the adaptability advantages of deep reinforcement learning. This integrated design addresses the research gap identified in recent literature [1], [8] and contributes toward robust, network-aware adaptive traffic signal optimization.

IV. Results

The proposed network-aware reinforcement learning-based traffic signal control system was implemented and evaluated

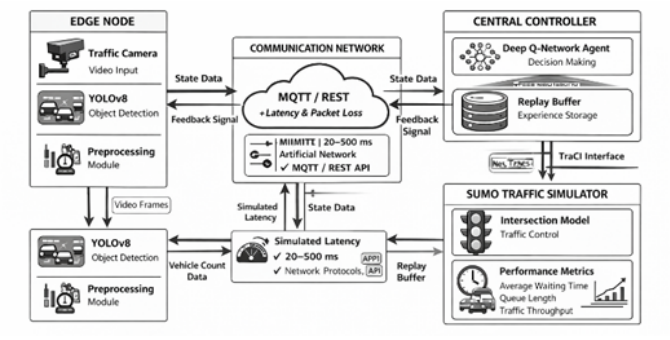

Fig. 1. Network-Aware Reinforcement Learning-Based Traffic Signal Control System model architecture.

using the SUMO (Simulation of Urban Mobility) traffic simulation environment, integrated with the TraCI (Traffic Control Interface) for real-time interaction between the simulation and the control agent. The system incorporates three key components: an edge-simulated perception module for vehicle detection, a communication layer introducing controlled network delays, and a centralized Deep Q-Network (DQN)-based reinforcement learning controller responsible for adaptive signal phase decisions. The evaluation focuses on assessing both traffic efficiency and robustness under varying communication conditions.

To ensure consistency with prior research, standard traffic performance metrics were used, including average vehicle waiting time, queue length, and traffic throughput. These metrics are widely adopted in reinforcement learning-based traffic signal control studies [1], [2], [9], as they directly reflect the effectiveness of signal timing strategies in reducing congestion and improving intersection efficiency. The performance of the proposed approach was compared against conventional fixed-time control and rule-based signal control methods, which serve as baseline approaches in traffic management systems.

The experimental results demonstrate that the reinforcement learning-based controller consistently outperforms traditional control strategies in dynamic traffic environments. In scenarios with varying traffic density, the DQN agent effectively learns adaptive policies that respond to real-time traffic conditions. Similar to findings reported in [9] and [11], the learned policy optimizes signal phase transitions by balancing queue reduction and traffic flow continuity. This results in a noticeable decrease in average waiting time and queue length compared to fixed-time scheduling, which operates based on static timing plans and fails to adapt to fluctuations in traffic demand.

The rule-based controller, while more flexible than fixed-time methods, still relies on predefined heuristics that do not generalize well across different traffic patterns. As a result, its performance is inferior to the reinforcement learning approach, particularly in scenarios involving sudden changes in traffic density. The DQN-based system, through continuous interaction with the environment, learns to anticipate congestion buildup and adjust signal phases accordingly, leading to improved throughput and smoother traffic flow. This behavior aligns with existing literature that highlights the superiority of deep reinforcement learning in adaptive traffic signal control [1], [4].

A key aspect of this study is the incorporation of communication delay modeling within the traffic control loop. Unlike most existing reinforcement learning-based traffic signal control systems, which assume instantaneous and reliable communication, the proposed framework introduces realistic network conditions by simulating latency and potential transmission delays. Multiple network scenarios were considered, including ideal conditions (no delay), moderate delay, and high-delay environments, to evaluate the robustness of the learned policy.

Under ideal network conditions, the reinforcement learning controller achieves optimal performance, with minimal waiting time and efficient traffic flow. This is expected, as the agent receives accurate and timely traffic state information, enabling precise decision-making. However, such conditions are rarely achievable in real-world intelligent transportation systems, where communication delays and uncertainties are inevitable.

When moderate communication delay is introduced, the system exhibits only a slight degradation in performance. The increase in waiting time and queue length is marginal, and the overall traffic throughput remains relatively stable. This indicates that the reinforcement learning agent has learned to operate effectively under delayed information conditions. Similar observations have been reported in robustness studies such as [8], where reinforcement learning models trained with environmental uncertainties demonstrate resilience to moderate disruptions. The ability of the proposed system to maintain stable performance under these conditions highlights the benefit of incorporating delay modeling during training, as it enables the agent to develop policies that are less sensitive to imperfect information.

In contrast, under high-delay scenarios, the performance of the system degrades more noticeably. Significant increases in average waiting time and queue length are observed, along with a reduction in overall traffic throughput. This degradation can be attributed to the delayed availability of traffic state information, which causes the controller to make decisions based on outdated observations. As a result, signal phase transitions may no longer align with the current traffic conditions, leading to inefficient traffic management. These findings are consistent with recent research emphasizing the importance of communication-aware control in intelligent transportation systems [12], [13].

Despite this degradation under extreme delay conditions, the proposed reinforcement learning-based system still demonstrates better adaptability compared to traditional fixed-time and rule-based approaches. While conventional methods are unaffected by communication delays due to their static or heuristic nature, they lack the ability to optimize traffic flow dynamically. In contrast, the reinforcement learning agent, even when operating under delayed information, continues to make context-aware decisions that partially mitigate congestion. This highlights the trade-off between adaptability and sensitivity to communication constraints.

Another important observation from the experimental results is the impact of delay-aware training on policy robustness. By exposing the reinforcement learning agent to delayed and imperfect state information during training, the model learns to generalize better across different network conditions. This approach addresses a key limitation identified in previous studies [8], where policies trained under ideal conditions fail to perform reliably in realistic environments. The proposed system demonstrates that incorporating communication constraints into the learning process can significantly enhance the practical applicability of reinforcement learning-based traffic control systems.

Furthermore, the integration of edge-simulated perception using YOLOv8 contributes to a more realistic system design. While the object detection module itself is not the primary focus of the study, it provides a practical mechanism for extracting traffic state information from visual data, similar to real-world deployments involving camera-based traffic monitoring systems. The use of edge-based processing ensures that raw video data is converted into compact traffic state representations before transmission, reducing communication overhead and aligning with emerging trends in edge computing for intelligent transportation systems [12].

Overall, the results validate that the proposed network-aware reinforcement learning framework achieves improved traffic signal performance compared to traditional control methods while maintaining robustness under realistic communication conditions. The system demonstrates the ability to adapt to dynamic traffic environments, reduce congestion, and sustain performance under moderate network delays. At the same time, the observed performance degradation under high-delay scenarios highlights the importance of considering communication constraints in the design and deployment of intelligent traffic control systems.

In conclusion, the experimental findings confirm that integrating reinforcement learning with network-aware design principles provides a more realistic and robust approach to traffic signal control. The proposed framework bridges the gap between simulation-based optimization and real-world deployment considerations, as identified in recent literature [1], [8], [12]. By explicitly modeling communication delays and incorporating them into the learning process, the system advances toward more practical and deployable intelligent transportation solutions.

V. Future Work

While the proposed network-aware reinforcement learning-based traffic signal control framework demonstrates improved adaptability and robustness under simulated network conditions, several extensions can be explored to enhance its scalability, realism, and practical deployment potential.

One important direction for future work is the extension from a single-intersection control system to a multi-intersection or network-level traffic control framework. In real-world urban environments, traffic flow at one intersection is strongly influenced by neighboring intersections. Integrating multi-agent reinforcement learning (MARL) approaches, as discussed in [5], [6], and [15], would enable coordination among multiple traffic signals to optimize global traffic efficiency. Such a distributed learning framework can reduce congestion propagation across the network and improve overall traffic throughput.

Another significant improvement involves exploring advanced reinforcement learning algorithms beyond Deep Q-Networks (DQN). While DQN is effective for discrete action spaces, policy-based and actor-critic methods such as Proximal Policy Optimization (PPO) or Advantage Actor-Critic (A2C) may offer better stability and faster convergence in complex traffic scenarios. Additionally, hierarchical reinforcement learning approaches [3] can be investigated to decompose traffic signal control into higher-level decision-making and lower-level signal timing optimization, thereby improving scalability and learning efficiency.

The current system models communication delay in a controlled and simplified manner. Future work can incorporate more realistic and dynamic network conditions, including variable latency distributions, packet loss, bandwidth limitations, and network congestion effects. Integrating real-world communication datasets or network simulators can provide a more accurate representation of intelligent transportation system environments. This would further strengthen the robustness of the learning model and provide deeper insights into the interaction between communication constraints and traffic control performance.

Another potential enhancement lies in improving the perception module. Although YOLOv8-based vehicle detection provides a practical approach for extracting traffic density information, future work can focus on handling real-world challenges such as occlusion, varying lighting conditions, and camera noise. Incorporating multi-modal sensing, such as combining vision-based detection with IoT sensors or radar data, can improve the accuracy and reliability of traffic state estimation. Additionally, optimizing the edge processing pipeline for real-time performance and resource efficiency would be critical for deployment on actual edge devices.

Federated and distributed learning frameworks also present a promising direction for future research. As highlighted in [7], federated deep reinforcement learning enables multiple intersections to collaboratively learn optimal policies without sharing raw data, thereby preserving privacy and reducing communication overhead. Integrating federated learning with network-aware traffic control can improve scalability and enable deployment across large urban traffic networks.

Furthermore, future work can explore the integration of predictive modeling techniques to enhance decision-making. By incorporating traffic flow prediction models, the reinforcement learning agent can anticipate future traffic conditions rather than relying solely on current observations. This predictive capability can improve the timing of signal phase transitions and further reduce congestion.

Finally, real-world deployment and validation remain critical steps for future research. The current study is based on sim-

ulation using the SUMO environment; therefore, implementing the proposed framework in a real-world or hardware-in-the-loop setup would provide valuable insights into its practical feasibility. Challenges such as hardware constraints, real-time processing requirements, and integration with existing traffic infrastructure must be addressed to move toward deployment-ready intelligent traffic systems.

In summary, future work will focus on enhancing scalability through multi-agent coordination, improving learning efficiency with advanced reinforcement learning techniques, incorporating realistic network and sensing conditions, and enabling practical deployment through distributed and real-world implementations. These directions will further strengthen the applicability of network-aware reinforcement learning in intelligent transportation systems.

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